VoteOnline - Secure Online Voting System

INTERNET PROGRAMMING LAB

AIM

To develop a secure and user-friendly e-voting website using HTML, CSS and JavaScript in which a voter logs in with the Voter ID, verifies two one-time passwords (at login and before voting), selects a candidate of his/her ward and submits one vote, while the administrator creates candidates, allocates election data and wards, creates and imports the voter list, updates it, and obtains the election results.

ALGORITHM

Step 1: Start. Load the election details, candidates, voter list and anonymous ballots from localStorage (seed demo data on the first run).

Step 2: Voter login: read the Voter ID and search it in the voter list; if it is not found, display an error message.

Step 3: OTP login verification: generate a random 6-digit OTP, display it in a simulated SMS box and compare it with the value typed by the voter. Allow access only when both match.

Step 4: Display the election name, date and status together with the voter details. If the voter has already voted or the election is closed, show a message and stop.

Step 5: Show only the candidates allocated to the voter's ward (name, party and symbol). The voter selects one candidate; the Get OTP for Voting button is then enabled.

Step 6: Generate a second OTP for voting and verify it. If it is correct, add the ballot (candidate ID and ward only, without the voter ID) to the ballot list and mark the voter as Voted, so that the voter cannot vote twice.

Step 7: Display the vote confirmation receipt with a confirmation number.

Step 8: Admin login (admin / admin123). Create an election candidate and allocate the candidate to a ward; set the election name, date and open / closed status.

Step 9: Admin - Voters: create a voter, import many voters from CSV text (VoterID, Name, Ward) skipping duplicates, view the list, change the ward of a voter and delete a voter.

Step 10: Admin - Results: count the ballots for each candidate ward-wise, sort them, display the counts with bars and mark the leading candidate.

Step 11: Save every change to localStorage. Logout returns to the home page. Stop.

PROGRAM

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1">
<title>VoteOnline - Secure E-Voting</title>
<link rel="stylesheet" href="style.css">
</head>
<body>
<header><div class="logo">&#128499; VoteOnline <small>Secure &bull; Transparent &bull; Accessible</small></div>
  <div><span id="who"></span> <button id="logout" class="btn light hidden">Logout</button></div></header>
<main>
<!-- LOGIN -->
<section id="login" class="two">
  <div class="card"><h2>Voter Login</h2>
    <label>Voter ID<input id="vid" placeholder="e.g. VTR1001"></label>
    <button id="vLogin" class="btn">Login &amp; Send OTP</button>
    <p class="hint">Demo voter IDs: VTR1001, VTR1002, VTR1003</p><p id="vMsg" class="msg err"></p></div>
  <div class="card"><h2>Admin Login</h2>
    <label>Username<input id="aUser"></label><label>Password<input id="aPass" type="password"></label>
    <button id="aLogin" class="btn alt">Admin Login</button>
    <p class="hint">Demo: admin / admin123</p><p id="aMsg" class="msg err"></p></div>
</section>

<!-- OTP -->
<section id="otpView" class="card narrow hidden">
  <h2 id="otpTitle">OTP Verification</h2>
  <div class="sms" id="sms"></div>
  <label>Enter 6-digit OTP<input id="otpIn" maxlength="6"></label>
  <button id="otpBtn" class="btn">Verify OTP</button> <button id="otpCancel" class="btn light2">Cancel</button>
  <p id="otpMsg" class="msg err"></p>
</section>

<!-- VOTER -->
<section id="voterView" class="hidden">
  <div class="card"><h2 id="elTitle"></h2><p id="voterInfo"></p></div>
  <div class="card" id="ballot"></div>
  <div class="card hidden" id="receipt"></div>
</section>

<!-- ADMIN -->
<section id="adminView" class="hidden">
  <nav id="tabs"><button data-t="t1" class="active">Candidates &amp; Election</button><button data-t="t2">Voter
      List</button><button data-t="t3">Import Voters</button><button data-t="t4">Results</button></nav>
  <div id="t1" class="tab active">
    <div class="two">
      <div class="card"><h3>Create Election Candidate</h3>
        <label>Candidate Name<input id="cName"></label><label>Party / Symbol<input id="cParty"></label>
        <label>Allocate to Ward<select id="cWard"></select></label>
        <button id="addCand" class="btn">Add Candidate</button></div>
      <div class="card"><h3>Election Details</h3>
        <label>Election Name<input id="elName"></label><label>Election Date<input id="elDate"
            type="date"></label>
        <label>Status<select id="elOpen"><option value="1">Open for voting</option><option
            value="0">Closed</option></select></label>
        <button id="saveEl" class="btn alt">Allocate Election Data</button></div>
    </div>
    <p id="a1Msg" class="msg"></p>
    <div class="card"><h3>Candidates</h3><div id="candTable"></div></div>
  </div>
  <div id="t2" class="tab">
    <div class="card"><h3>Create Voter</h3>
      <div class="inline"><input id="nvId" placeholder="Voter ID"><input id="nvName" placeholder="Voter
          name"><select id="nvWard"></select><button id="addVoter" class="btn">Add Voter</button></div>
      <p id="a2Msg" class="msg"></p></div>
    <div class="card"><h3>View &amp; Update Voter List (allocate users to wards)</h3><div
        id="voterTable"></div></div>
  </div>
  <div id="t3" class="tab">
    <div class="card"><h3>Import Voter List (CSV: VoterID,Name,Ward)</h3>
      <textarea id="csv" rows="6" placeholder="VTR2001,Anand,Ward 1&#10;VTR2002,Bhavani,Ward 2"></textarea>
      <button id="importBtn" class="btn">Import Voters</button><p id="a3Msg" class="msg"></p></div>
  </div>
  <div id="t4" class="tab"><div class="card"><h3>Election Results</h3><div id="results"></div></div></div>
</section>
</main>
<script src="script.js"></script>
</body>
</html>
```

style.css

```
:root{--g:#0b6e4f;--navy:#12355b;--bg:#eef5f1}
*{box-sizing:border-box}
body{margin:0;font-family:"Segoe UI",Arial,sans-serif;background:var(--bg);color:#1d2b26}
header{display:flex;justify-content:space-between;align-items:center;background:linear-
    gradient(90deg,#12355b,#0b6e4f);color:#fff;padding:12px 26px}
.logo{font-size:24px;font-weight:700}.logo small{font-size:12px;font-weight:400;margin-left:8px;color:#bfe6d8}
main{max-width:1000px;margin:18px auto;padding:0 14px}
.card{background:#fff;border-radius:10px;padding:16px;margin-bottom:14px;box-shadow:0 1px 5px
    rgba(18,53,91,.15)}
.narrow{max-width:420px;margin:30px auto}
.two{display:grid;grid-template-columns:1fr 1fr;gap:14px}@media(max-width:760px){.two{grid-template-
    columns:1fr}}
h2,h3{margin-top:0;color:var(--navy)}
label{display:flex;flex-direction:column;font-size:13px;font-weight:600;margin-bottom:10px;gap:3px}
input,select,textarea{padding:8px;border:1px solid #b5c9c0;border-radius:6px;font:inherit}
textarea{width:100%;margin-bottom:8px}
.btn{background:var(--g);color:#fff;border:0;border-radius:6px;padding:8px 16px;cursor:pointer;font:inherit}
.btn.alt{background:var(--navy)}.btn.light{background:#fff;color:var(--
    navy)}.btn.light2{background:#dfe8e4;color:#234}.btn.red{background:#c0392b}.btn.sm{padding:4px 10px;font-
    size:12px}
.btn:disabled{background:#aaa;cursor:not-allowed}
.hint{font-size:12px;color:#566}.msg{font-weight:700;color:#1e7d3a;min-height:18px}.msg.err{color:#c0392b}
.hidden{display:none!important}
.sms{background:#fffbe6;border:1px dashed #d4a017;border-radius:8px;padding:10px;margin-bottom:12px;font-
    size:13px}
.cand{display:flex;align-items:center;gap:12px;border:2px solid #cfe1d8;border-radius:10px;padding:10px
    14px;margin:8px 0;cursor:pointer}
.cand.sel{border-color:var(--g);background:#e6f6ee}
.cand .sym{font-size:34px}.cand small{color:#566}
nav{display:flex;gap:4px;margin-bottom:12px}nav button{border:0;background:#d8e6df;padding:8px 16px;border-
    radius:6px;cursor:pointer;font:inherit}nav button.active{background:var(--g);color:#fff}
.tab{display:none}.tab.active{display:block}
.inline{display:flex;gap:8px;flex-wrap:wrap}.inline input,.inline select{flex:1;min-width:140px}
table{width:100%;border-collapse:collapse;font-size:13px}th,td{padding:7px;border-bottom:1px solid #dbe7e1;text-
    align:left}th{background:#e0efe8}
.bar{height:18px;background:var(--g);border-radius:4px;min-width:2px}
.badge{padding:2px 8px;border-radius:10px;color:#fff;font-
    size:11px}.yes{background:#1e7d3a}.no{background:#d98a00}
.receipt{border:2px solid var(--g);border-radius:10px;padding:14px;background:#f1fbf6}
```

script.js

```
const $ = id => document.getElementById(id);
const WARDS = ['Ward 1','Ward 2','Ward 3'];
const seed = () => ({
  election:{name:'Municipal Council Election 2026', date:'2026-11-15', open:true},
  candidates:[{id:1,name:'Anitha Rajan',party:'Green Alliance',ward:'Ward 1',icon:'\u{1F33F}'},
              {id:2,name:'Mohan Das',party:'People First',ward:'Ward 1',icon:'\u{2B50}'},
              {id:3,name:'Farida Begum',party:'Unity Front',ward:'Ward 1',icon:'\u{1F54A}'},
              {id:4,name:'Suresh Kumar',party:'Green Alliance',ward:'Ward 2',icon:'\u{1F33F}'},
              {id:5,name:'Latha Devi',party:'People First',ward:'Ward 2',icon:'\u{2B50}'}],
  voters:[{id:'VTR1001',name:'Arun Prakash',ward:'Ward 1',voted:false},
          {id:'VTR1002',name:'Divya Menon',ward:'Ward 1',voted:false},
          {id:'VTR1003',name:'Karthik Raja',ward:'Ward 2',voted:false}],
  votes:[{cand:1,ward:'Ward 1'},{cand:1,ward:'Ward 1'},{cand:2,ward:'Ward 1'},{cand:4,ward:'Ward 2'}],  //
      anonymous ballots
  nextCand:6});
let db = JSON.parse(localStorage.getItem('voteonline') || 'null') || seed();
const save = () => localStorage.setItem('voteonline', JSON.stringify(db));
let voter = null, isAdmin = false, otp = null, otpPurpose = '', chosen = null;
const say = (el, t, bad) => { $(el).textContent = t; $(el).className = 'msg' + (bad ? ' err' : ''); };
const show = v => ['login','otpView','voterView','adminView'].forEach(s => $(s).classList.toggle('hidden', s !==
     v));

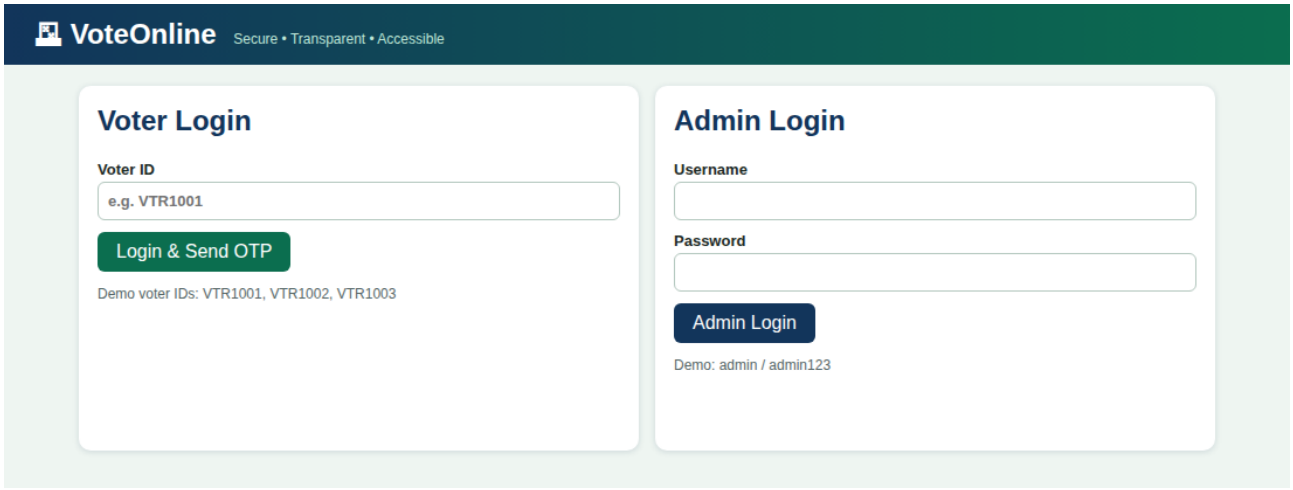
/* ---------- OTP helper ---------- */
function sendOtp(purpose){
  otp = String(Math.floor(100000 + Math.random()*900000)); otpPurpose = purpose;
  $('otpTitle').textContent = purpose === 'login' ? 'Login OTP Verification' : 'OTP for Casting Vote';
  $('sms').innerHTML = `&#128241; <b>SMS to registered mobile of ${voter.id}:</b> Your VoteOnline OTP is <b
      id="otpVal">${otp}</b>. Do not share it. (demo display)`;
  $('otpIn').value = ''; $('otpMsg').textContent = ''; show('otpView');
}
$('otpBtn').onclick = () => {
  if($('otpIn').value.trim() !== otp){ return say('otpMsg','Wrong OTP. Try again.',true); }
  if(otpPurpose === 'login'){ renderVoter(); show('voterView'); }
  else { castVote(); }
};
$('otpCancel').onclick = () => { if(otpPurpose === 'login'){ voter = null; show('login'); } else
    show('voterView'); };

/* ---------- login / logout ---------- */
$('vLogin').onclick = () => {
  const v = db.voters.find(x => x.id === $('vid').value.trim().toUpperCase());
  if(!v) return say('vMsg','Voter ID not found in the voter list.',true);
  voter = v; $('who').textContent = v.name; $('logout').classList.remove('hidden'); sendOtp('login');
};
$('aLogin').onclick = () => {
  if($('aUser').value !== 'admin' || $('aPass').value !== 'admin123') return say('aMsg','Invalid admin
      credentials.',true);
  isAdmin = true; $('who').textContent = 'Administrator'; $('logout').classList.remove('hidden'); renderAdmin();
       show('adminView');
};
$('logout').onclick = () => { voter = null; isAdmin = false; $('who').textContent = '';
    $('logout').classList.add('hidden'); $('vid').value = ''; show('login'); };

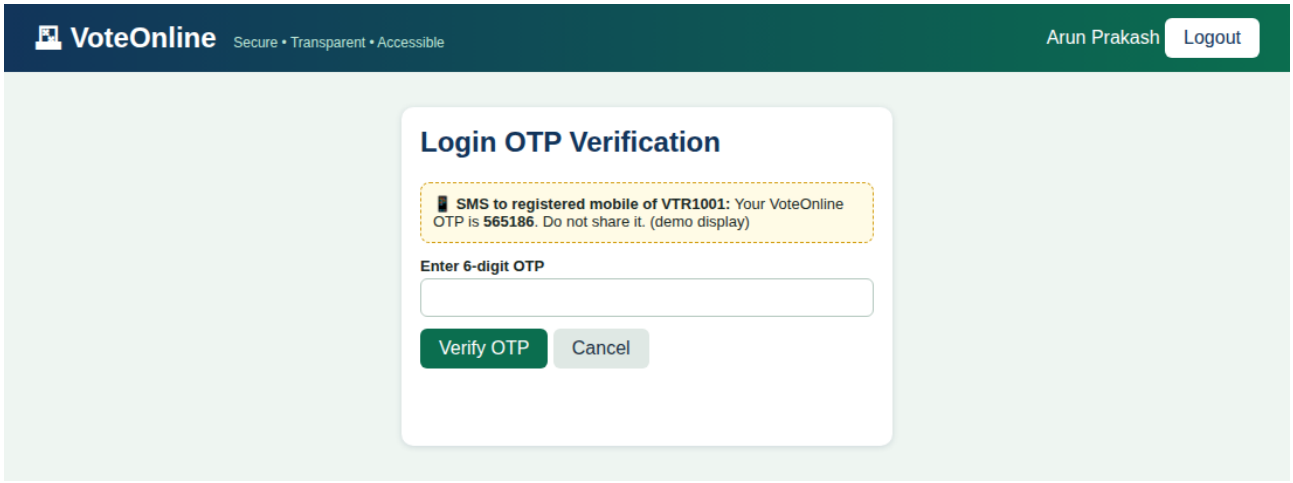
/* ---------- voter ---------- */
function renderVoter(){
  const v = db.voters.find(x => x.id === voter.id), e = db.election;
  $('elTitle').textContent = e.name; $('receipt').classList.add('hidden'); chosen = null;
  $('voterInfo').innerHTML = `Voter: <b>${v.name}</b> (${v.id})  |  Ward: <b>${v.ward}</b>
       |  Election date: ${e.date}  |  Status: <b>${e.open ? 'Open' : 'Closed'}</b>`;
  if(v.voted) { $('ballot').innerHTML = '<h3>You have already cast your vote.</h3><p>Thank you for
      participating. A voter can vote only once.</p>'; return; }
  if(!e.open) { $('ballot').innerHTML = '<h3>Voting is closed.</h3>'; return; }
  const list = db.candidates.filter(c => c.ward === v.ward);
  $('ballot').innerHTML = `<h3>Select your candidate</h3>` + list.map(c => `<div class="cand" data-id="${c.id}"
      onclick="choose(${c.id})">
    <span class="sym">${c.icon}</span><div><b>${c.name}</b><br><small>${c.party}</small></div></div>`).join('')
        +
    `<button id="getOtp" class="btn" disabled onclick="sendOtp('vote')">Get OTP for Voting</button> <span
        class="hint">Select a candidate to continue</span>`;
}
function choose(id){ chosen = id; document.querySelectorAll('.cand').forEach(d => d.classList.toggle('sel',
    +d.dataset.id === id)); $('getOtp').disabled = false; }
function castVote(){
  const v = db.voters.find(x => x.id === voter.id), c = db.candidates.find(x => x.id === chosen);
  if(v.voted) return;
  db.votes.push({cand:c.id, ward:v.ward}); v.voted = true; save();       // ballot stored without voter identity
  show('voterView'); renderVoter();
  $('receipt').classList.remove('hidden');
  $('receipt').innerHTML = `<div class="receipt"><h3>&#9989; Vote submitted successfully</h3>
    <p>Confirmation No: <b>VOL-${Date.now().toString().slice(-8)}</b><br>Voter ID: ${v.id}   Ward:
        ${v.ward}<br>Your vote has been recorded securely and anonymously.</p></div>`;
}
```

```
/* ---------- admin ---------- */
document.querySelectorAll('#tabs button').forEach(b => b.onclick = () => {
  document.querySelectorAll('#tabs button').forEach(x => x.classList.toggle('active', x === b));
  document.querySelectorAll('.tab').forEach(t => t.classList.toggle('active', t.id === b.dataset.t));
  renderAdmin(); });
const wardOpts = sel => WARDS.map(w => `<option ${w === sel ? 'selected' : ''}>${w}</option>`).join('');
$('addCand').onclick = () => {
  if(!$('cName').value.trim()) return say('a1Msg','Enter candidate name.',true);
  db.candidates.push({id:db.nextCand++, name:$('cName').value, party:$('cParty').value || 'Independent',
      ward:$('cWard').value, icon:'\u{1F5F3}'});
  save(); $('cName').value = $('cParty').value = ''; say('a1Msg','Candidate created and allocated to ' +
      $('cWard').value + '.'); renderAdmin();
};
$('saveEl').onclick = () => { db.election = {name:$('elName').value, date:$('elDate').value,
    open:$('elOpen').value === '1'}; save(); say('a1Msg','Election data allocated and saved.'); };
$('addVoter').onclick = () => {
  const id = $('nvId').value.trim().toUpperCase(); if(!id || !$('nvName').value) return say('a2Msg','Enter voter
       ID and name.',true);
  if(db.voters.some(v => v.id === id)) return say('a2Msg','Voter ID already exists.',true);
  db.voters.push({id, name:$('nvName').value, ward:$('nvWard').value, voted:false}); save(); $('nvId').value =
      $('nvName').value = ''; say('a2Msg','Voter added.'); renderAdmin();
};
$('importBtn').onclick = () => {
  let n = 0, skip = 0;
  $('csv').value.split('\n').forEach(l => { const [id, name, ward] = l.split(',').map(s => s && s.trim());
    if(!id || !name) return; if(db.voters.some(v => v.id === id.toUpperCase())) { skip++; return; }
    db.voters.push({id:id.toUpperCase(), name, ward:WARDS.includes(ward) ? ward : 'Ward 1', voted:false}); n++;
      });
  save(); say('a3Msg', `${n} voter(s) imported, ${skip} duplicate(s) skipped.`); renderAdmin();
};
function setWard(id){ db.voters.find(v => v.id === id).ward = $('w' + id).value; save(); renderAdmin(); }
function delVoter(id){ db.voters = db.voters.filter(v => v.id !== id); save(); renderAdmin(); }
function delCand(id){ db.candidates = db.candidates.filter(c => c.id !== id); save(); renderAdmin(); }
function renderAdmin(){
  ['cWard','nvWard'].forEach(id => { const cur = $(id).value; $(id).innerHTML = wardOpts(cur); });
  $('elName').value = db.election.name; $('elDate').value = db.election.date; $('elOpen').value =
      db.election.open ? '1' : '0';
  $('candTable').innerHTML =
      '<table><tr><th>ID</th><th>Candidate</th><th>Party</th><th>Ward</th><th>Action</th></tr>' +
    db.candidates.map(c => `<tr><td>${c.id}</td><td>${c.icon}
        ${c.name}</td><td>${c.party}</td><td>${c.ward}</td><td><button class="btn sm red"
        onclick="delCand(${c.id})">Delete</button></td></tr>`).join('') + '</table>';
  $('voterTable').innerHTML = '<table><tr><th>Voter
      ID</th><th>Name</th><th>Ward</th><th>Voted?</th><th>Action</th></tr>' +
    db.voters.map(v => `<tr><td>${v.id}</td><td>${v.name}</td><td><select id="w${v.id}"
        onchange="setWard('${v.id}')">${wardOpts(v.ward)}</select></td>
    <td><span class="badge ${v.voted ? 'yes' : 'no'}">${v.voted ? 'Voted' : 'Not yet'}</span></td><td><button
        class="btn sm red" onclick="delVoter('${v.id}')">Delete</button></td></tr>`).join('') + '</table>';
  const total = db.votes.length, voted = db.voters.filter(v => v.voted).length;
  let h = `<p>Total votes cast: <b>${total}</b>  |  Registered voters:
      <b>${db.voters.length}</b></p>`;
  WARDS.forEach(w => {
    const cs = db.candidates.filter(c => c.ward === w); if(!cs.length) return;
    const tally = cs.map(c => ({c, n:db.votes.filter(v => v.cand === c.id).length})).sort((a, b) => b.n - a.n),
        max = Math.max(1, tally[0].n);
    h += `<h4>${w}</h4><table>` + tally.map((t, i) => `<tr><td width="30%">${t.c.name}
        <small>(${t.c.party})</small>${i === 0 && t.n ? ' &#127942;' : ''}</td>
      <td width="55%"><div class="bar" style="width:${t.n / max * 100}%"></div></td><td><b>${t.n}</b>
          votes</td></tr>`).join('') + '</table>';
  });
  $('results').innerHTML = h;
}
show('login');
```

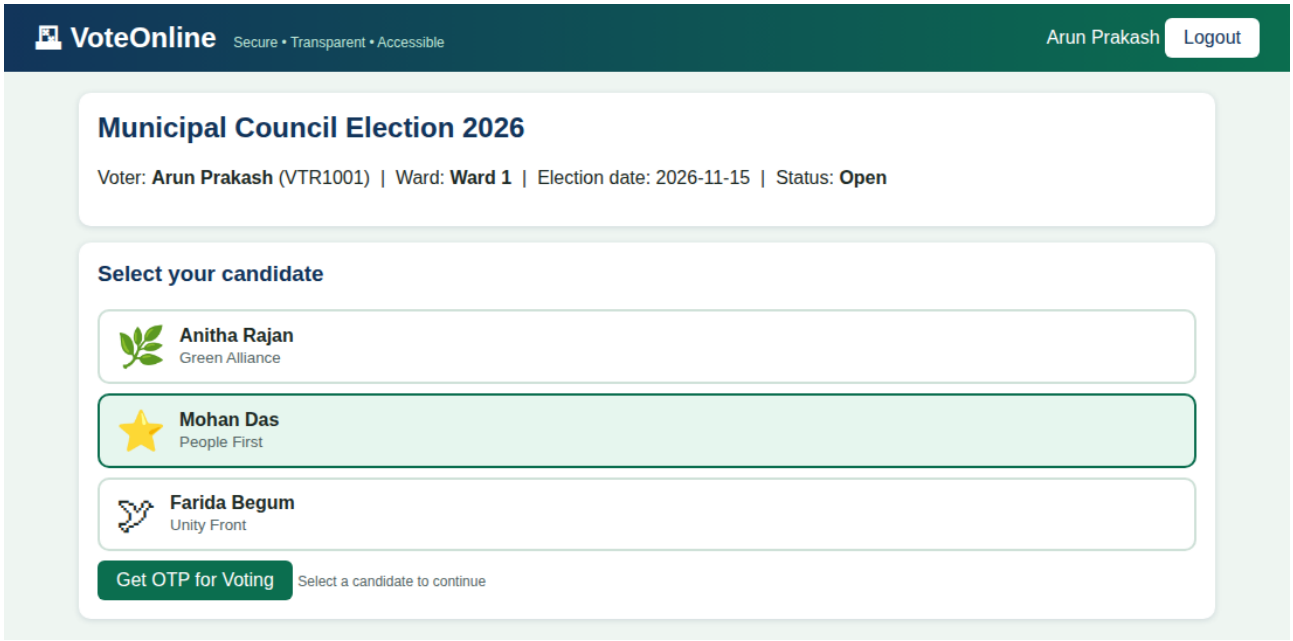
OUTPUT



Home page with Voter Login (Voter ID) and Admin Login



OTP login verification - the OTP is sent to the voter's registered mobile number



Ballot page showing only the candidates of the voter's ward; a candidate is selected

VoteOnline Secure • Transparent • Accessible

Arun Prakash Logout

Municipal Council Election 2026

Voter: **Arun Prakash** (VTR1001) | Ward: **Ward 1** | Election date: 2026-11-15 | Status: **Open**

You have already cast your vote.

Thank you for participating. A voter can vote only once.

✅ **Vote submitted successfully**

Confirmation No: **VOL-58975615**
Voter ID: VTR1001 Ward: Ward 1
Your vote has been recorded securely and anonymously.

Vote submitted - confirmation receipt; the voter cannot vote again

VoteOnline Secure • Transparent • Accessible

Administrator Logout

Candidates & Election | Voter List | Import Voters | Results

Create Election Candidate

Candidate Name

Party / Symbol

Allocate to Ward

Ward 2

Add Candidate

Election Details

Election Name

Municipal Council Election 2026

Election Date

11/15/2026

Status

Open for voting

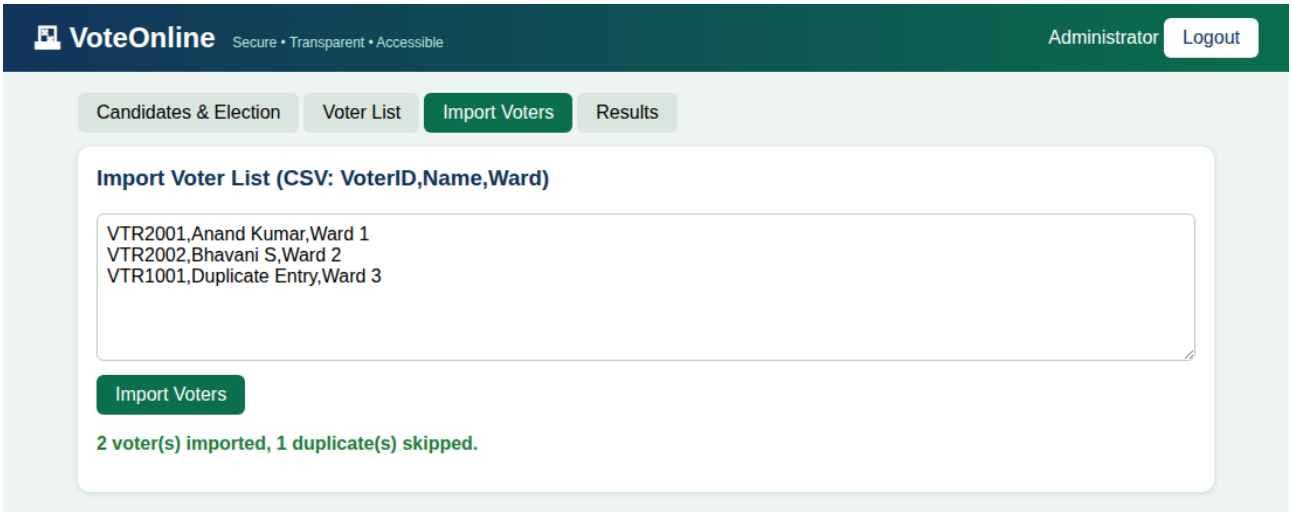
Allocate Election Data

Candidate created and allocated to Ward 2.

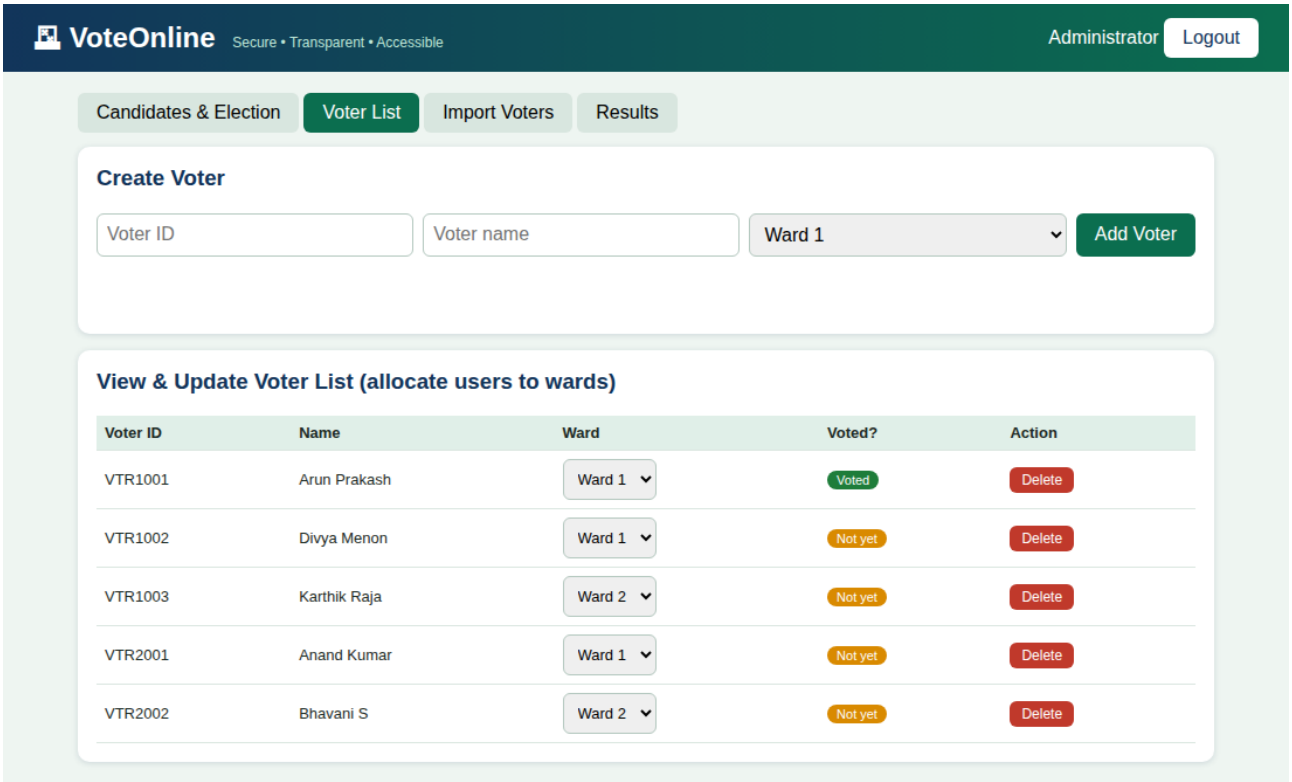
Candidates

ID	Candidate	Party	Ward	Action
1	🌿 Anitha Rajan	Green Alliance	Ward 1	Delete
2	⭐ Mohan Das	People First	Ward 1	Delete
3	🕊 Farida Begum	Unity Front	Ward 1	Delete
4	🌿 Suresh Kumar	Green Alliance	Ward 2	Delete
5	⭐ Latha Devi	People First	Ward 2	Delete
6	🗳 Rekha Nair	Unity Front	Ward 2	Delete

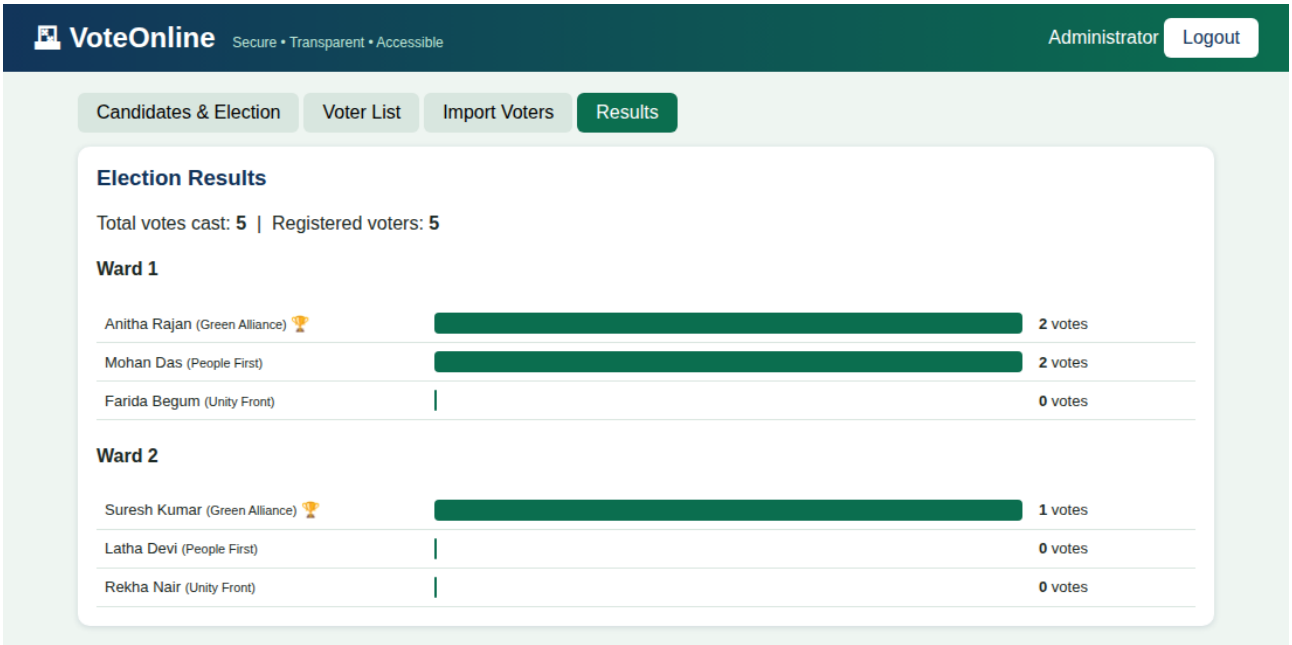
Admin: create election candidate, allocate candidate to a ward and set election details



Admin: importing the voter list from CSV data (one duplicate voter ID is skipped)



Admin: view and update the voter list - allocate voters to wards, see voting status, delete voters



Admin: election results ward-wise with vote counts

RESULT

The VoteOnline e-voting website was developed using HTML, CSS and JavaScript. Voters could log in with the Voter ID, verify the OTPs, choose a candidate and submit a single vote, and the administrator could manage candidates, wards and voter lists and obtain the election results. The output was verified successfully.